

Engineering Report

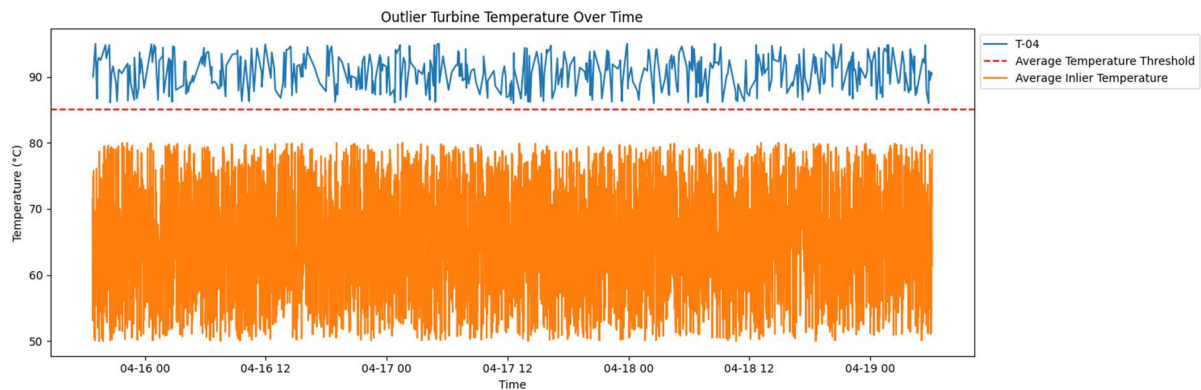
By
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Report to summarise my findings and suggestions for a new data pipeline design to better accommodate the surge in telemetry data being received from AeroGrid wind turbines

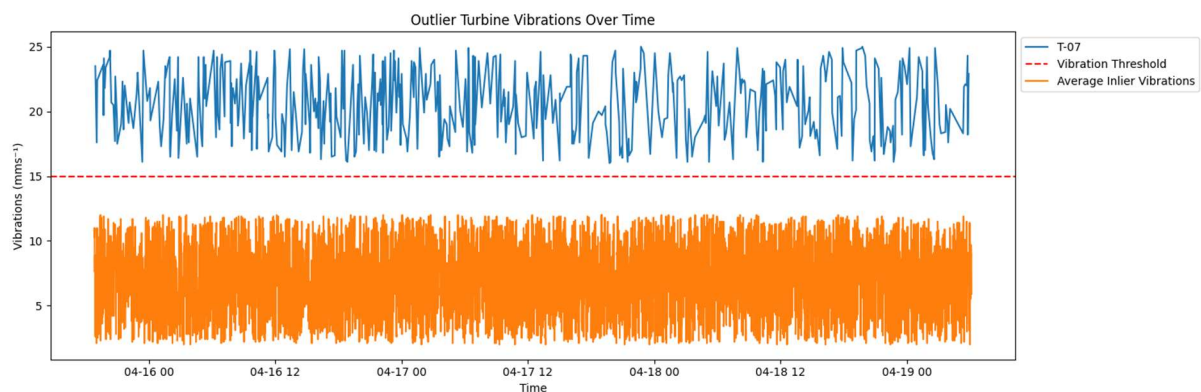
Findings

To identify which turbines were failing, I developed the python script 'turbine_analysis.py'. In this script I parsed the telemetry data given to me and identified which turbines are failing the anomaly rules (average temperature cannot exceed 85°C and the vibration levels cannot spike above 15mm/s⁻¹). From the analysis done by the script, I can determine that there are 2 turbines which are failing; T-04 has an average temperature of 90.6 degrees, and T-07 is exceeding the vibration levels with an average vibration of 20.6 mm/s⁻¹.

To visualize these outliers, I compiled graphs [Fig 1 and Fig 2] to show how they compare to the other turbines and by how much they exceed the anomaly rules



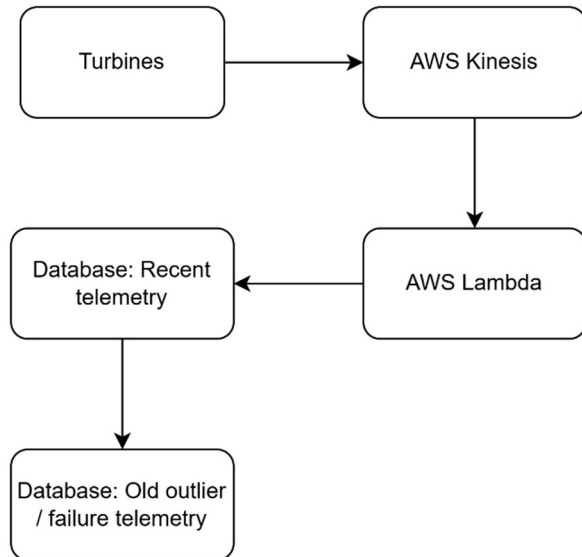
[Fig 1: T-04 Temperature over Time]



[Fig 2: T-07 Vibrations over Time]

Architecture Solution

To better accommodate the massive, non-stop flow of data, I propose a new data pipeline [Fig 3] which utilizes cloud infrastructure which can handle this data in real time.



[Fig 3: Data Pipeline Diagram]

Once the data leaves the turbines, it will go to the AWS Kinesis where it will be held before it is processed. It will then go to AWS Lambda where it will be analysed to spot any outliers. Then it will be delivered to our server to be stored.

To optimise what data is stored on our server, an average of the data for each turbine could be stored every minute, rather than continuous. After a week or so, the data could be trimmed to only keep outlying data.